

PROJECT REPORT
ON
“ Fake News Detection System Using Machine Learning ”

Submitted in the partial fulfilment of
the requirement for the Award of Degree of
Bachelor of Technology
IN
Computer Science & Engineering



Session 2023 -2027
PROJECT CODE – BTCS 603-18

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Introduction

Fake News Detection is a Machine Learning-based system designed to identify whether a news article is genuine or fake. With the rapid growth of the internet and social media platforms, news spreads faster than ever before. However, along with authentic information, a large amount of false or misleading news is also shared online. Fake news can create confusion, influence public opinion, and lead to social, political, and economic consequences.

The primary objective of this project is to develop an automated system that can analyze news content and classify it as either "Real" or "Fake." The system uses Natural Language Processing (NLP) techniques and Machine Learning algorithms to process textual data and make accurate predictions. In this project, TF-IDF Vectorization is used to convert text into numerical features, and Logistic Regression is used as the classification model.

The Fake News Detection System helps users quickly verify the authenticity of news articles, reducing the spread of misinformation.

This project demonstrates the practical application of Machine Learning and Data Science in solving real-world problems and improving information reliability in the digital age.

Problem Statement

The rapid growth of social media, online news portals, and digital communication platforms has made it easy to share information instantly. However, this has also led to the widespread dissemination of fake news, misleading articles, and false information. Identifying the authenticity of news manually is challenging due to the large volume of content generated every day.

Fake news can negatively impact society by spreading rumors, influencing public opinion, creating panic, and causing social and political unrest. Traditional methods of fact-checking are timeconsuming and require significant human effort.

Therefore, there is a need for an automated system that can efficiently analyze news articles and determine whether the information is real or fake. The proposed Fake News Detection System uses Machine Learning and Natural Language Processing techniques to classify news content accurately, helping users verify information and reduce the spread of misinformation.

Objectives

- To develop a system that can detect fake and real news.
- To use Machine Learning techniques for news classification.
- To analyze news articles based on their content.
- To reduce the spread of false information.
- To provide users with reliable and accurate news verification.

Technology Stack

- **Programming Language:** Python
- **Machine Learning Library:** Scikit-learn
- **Data Handling:** Pandas, NumPy
- **Web Framework:** Flask
- **Frontend:** HTML, CSS
- **Development Environment:** Visual Studio Code (VS Code)
- **Dataset:** Fake.csv and True.csv

Dataset

The dataset used in this project consists of two files:

- **Fake.csv** – Contains fake news articles.
- **True.csv** – Contains real news articles.

These datasets include news titles, text content, and other related information. The data is used to train and test the Machine Learning model for detecting whether a news article is real or fake.

Data Preprocessing

Data preprocessing is the process of preparing the dataset before training the Machine Learning model. In this project, the Fake.csv and True.csv datasets are loaded and combined into a single dataset. Labels are assigned to identify fake and real news articles.

The news title and text are combined to create complete content.

Missing values and unnecessary characters are removed to make the data clean and consistent. Finally, the dataset is shuffled and prepared for feature extraction and model training.

Data preprocessing helps improve the accuracy and performance of the Fake

News Detection System. **Feature Extraction**

Feature Extraction is the process of converting text data into numerical values that a Machine Learning model can understand. Since computers cannot directly process text, the news content must be transformed into a suitable format.

In this project, **TF-IDF (Term Frequency-Inverse Document Frequency)** is used for feature extraction. TF-IDF measures the importance of words in a news article based on how frequently they appear in a document and across the dataset.

The extracted features help the model identify patterns and distinguish between fake and real news. This improves the accuracy of the Fake News Detection System.

Machine Learning Model

A Machine Learning model is used to classify news articles as either real or fake. In this project, **Logistic Regression** is used because it is simple, efficient, and provides good accuracy for text classification tasks.

The model is trained using the processed news dataset and TF-IDF features. During training, the model learns patterns and relationships between words and news categories. After training, it can predict whether a new news article is real or fake based on its content.

The Logistic Regression model helps the system make fast and accurate predictions, making it suitable for fake news detection.

System Architecture

The System Architecture of the Fake News Detection System consists of several steps. First, the user enters a news article through the web interface. The entered text is then processed and converted into numerical features using TF-IDF Vectorization.

These features are passed to the trained Logistic Regression model. The model analyzes the news content and predicts whether the news is real or fake. Finally, the prediction result is displayed to the user on the screen.

Flow:

User Input → Data Preprocessing → TF-IDF Vectorization → Logistic Regression Model → Prediction → Result Display

Implementation

The Fake News Detection System is implemented using Python, Flask, HTML, and CSS. The news dataset is first processed and converted into numerical features using TF-IDF Vectorization. A Logistic Regression model is then trained on the dataset to classify news as real or fake.

The Flask framework is used to create the backend of the application, while HTML and CSS are used to design the user interface. Users can enter a news article through the web page, and the trained model analyzes the content and predicts whether the news is real or fake. The prediction result is then displayed on the screen.

This implementation provides a simple, user-friendly, and efficient way to detect fake news.

Results

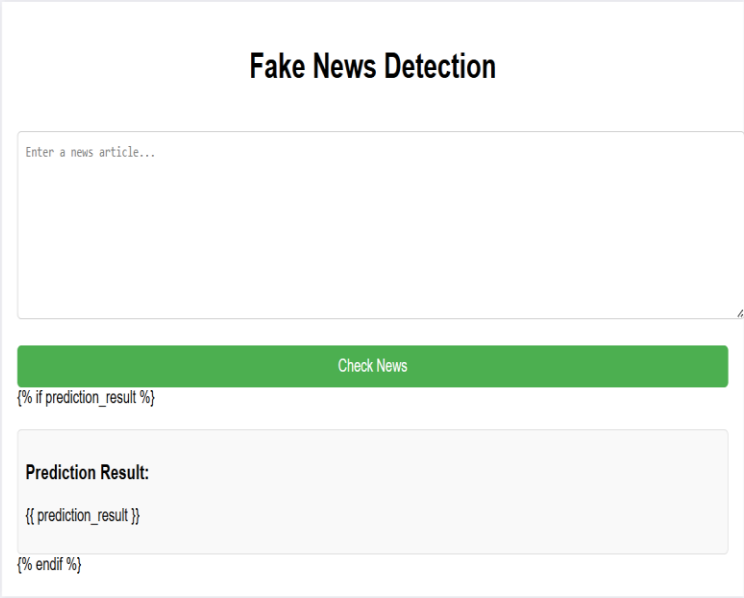
The Fake News Detection System was successfully developed and tested using a news dataset. The Logistic Regression model was able to classify news articles as real or fake based on their content. The system provided accurate predictions and demonstrated good performance in detecting fake news.

Users can enter a news article through the web interface and receive instant results. The project shows that Machine Learning techniques can effectively help in identifying misinformation and improving the reliability of online news.

Figure 1: Fake News Detection Web Application Interface

The above figure shows the user interface of the Fake News Detection System developed using Flask, HTML, and CSS. Users can enter a news article in the text area and click the "Check News" button to analyze the content. The system processes the input using TF-IDF Vectorization and the trained Logistic Regression model to predict whether the news is Real or Fake. The prediction result is then displayed on the screen, providing users with quick and reliable news verification.

- The system provides a simple and user-friendly interface for news verification.
- Users can easily enter news content and get instant prediction results.
- Machine Learning techniques help improve the accuracy of fake news detection.
- The web application reduces the effort required for manual fact-checking.
- This interface demonstrates the practical implementation of Machine Learning in solving real-world problems.



The screenshot displays a web application titled "Fake News Detection". It features a large text input area with the placeholder text "Enter a news article...". Below this input area is a green button labeled "Check News". Underneath the button, there is a conditional display block: "{% if prediction_result %}" followed by a box containing the text "Prediction Result:" and a Jinja2 template variable "{{ prediction_result }}", which is then closed with "{% endif %}".

Advantages

- Detects fake and real news automatically.
- Provides quick and accurate results.
- Reduces the spread of misinformation.
- Saves time compared to manual verification.
- Easy to use through a simple web interface.
- Supports decision-making with reliable information.
- Can be applied to various online news platforms.

Future Scope

The Fake News Detection System can be improved in the future by using advanced Machine Learning and Deep Learning techniques to achieve higher accuracy. The system can be integrated with social media platforms and news websites for real-time fake news detection.

Future enhancements may include support for multiple languages, image and video verification, and mobile application development. These improvements will make the system more efficient, reliable, and useful for a larger number of users.

Conclusion & Questions

The Fake News Detection System successfully uses Machine Learning techniques to classify news articles as real or fake. By applying data preprocessing, TF-IDF feature extraction, and Logistic Regression, the system provides accurate and reliable predictions.

This project demonstrates how Machine Learning can help reduce the spread of misinformation and support users in identifying trustworthy news. The system is simple, efficient, and can be further enhanced with advanced technologies in the future.

Thank You